

Repository 3

Agentic delegation and the language frontier

Quispe & Xu (2026)

github.com/davidolano03/ai-03-quispe

20 minutes

1. The citation check changes the object

	Prompt citation	arXiv 2605.25438v2
Title	<i>Coding Beyond Your Training: Claude Code and the Technological Frontier of Software Developers</i>	<i>Agentic Delegation and the Language Frontier of Software Developers: A Model and Evidence from Claude Code on GitHub</i>
Authors	Quispe	Alexander Quispe; Kevin Xu
Date	2026 only	v1: May 25; v2: July 7; PDF: July 8, 2026
Sample	not stated	5,346 developers; 28 months; 3.2m commits; 57m changed files

The old title is real—the v2 PDF identifies it as an earlier circulated version—but it is not the current citation and it omits Xu.

Source: Quispe and Xu, arXiv:2605.25438v2, record, title page, abstract, and footnote 1.

2. What frontier is expanding?

Production frontier

The set of languages in which a developer can **ship code now**.

- ▶ familiar set $\mathcal{K}_i$
- ▶ unfamiliar set $\mathcal{U}_i$
- ▶ active count $N_{it}^g = \sum_k Z_{ik,t}^g$

Not necessarily learning

Delegation may activate a language without raising the developer's language-specific skill s .

- ▶ developer specifies and verifies
- ▶ agent executes a share λ
- ▶ learning is a later, ambiguous signal channel

Source: Quispe and Xu, Section 3 and Appendix learning extension.

3. Three modes, one opportunity shock

$$V^S = \omega + s\mu - \frac{\rho s^2}{2\pi} - b,$$

$$V^C = V^S + \gamma s - r_C,$$

$$V^D = \omega + (1 - \lambda)s\mu + \lambda az(A) - \kappa(a, s) - r_D - b \\ - \frac{\rho}{2} \left[\frac{(1 - \lambda)^2 s^2}{\pi} + \sigma_D^2(a, s, A) \right].$$

Solo
skill + match risk

Conversation
gain scales with s

Delegation
agent output + verification

Source: Quispe and Xu, Equations (1)–(3).

4. Entry is a threshold comparison

$$T^S = b - s\mu + \frac{\rho s^2}{2\pi},$$

$$T^C = T^S - (\gamma s - r_C),$$

$$T^1 = \min\{T^S, T^C\} = T^S - \max\{0, \gamma s - r_C\},$$

$$T^D = b - (1 - \lambda)s\mu - \lambda az(A) + \kappa(a, s) + r_D + \frac{\rho}{2} \left[\frac{(1 - \lambda)^2 s^2}{\pi} + \sigma_D^2 \right],$$

$$T^2 = \min\{T^1, T^D\} \leq T^1.$$

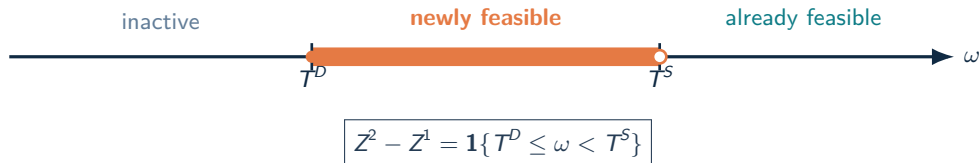
A language is active iff $\omega \geq T^g$. Menu nesting, $\{S, C\} \subset \{S, C, D\}$, mechanically proves $Z^2 \geq Z^1$ and $N^2 \geq N^1$ path by path.

Source: Quispe and Xu, Equations (4)–(7), Proposition 1.

5. The main object: the activation band

Conditions. Unfamiliar language; Assumption 1 $\gamma s - r_C \leq 0$, hence $T^1 = T^S$; and

$$B \equiv T^S - T^D = \lambda[az(A) - s\mu] - \kappa - r_D + \frac{\rho}{2} \left[\frac{(2\lambda - \lambda^2)s^2}{\pi} - \sigma_D^2 \right] > 0.$$



Source: Quispe and Xu, Equation (8), Proposition 2.

6. Conditions are doing real work

Condition	What it buys	If it fails
Assumption 1: $\gamma s - r_C \leq 0$ $B = T^S - T^D > 0$ activity means $V \geq 0$ F continuous opportunity mass in band	old unfamiliar-language threshold is T^S nonempty strict activation band exact endpoints $[T^D, T^S)$ probability $F(T^S) - F(T^D)$ positive average effect	compare T^D with T^1 , not necessarily T^S $B = 0$: empty; $B < 0$: old threshold binds changing the convention changes boundary cases atoms require left-limit bookkeeping pathwise weak expansion can coexist with zero mean change

At $\omega = T^D$: new menu active, old menu inactive. At $\omega = T^S$: both menus active.

Source: Quispe and Xu, Proposition 2 and proof.

7. Hand derivation: checking the activation band

Derivación a mano:

Prop. 2:

Excedente de trabajar solo: $V^S = \omega + sH - \frac{\beta s^2}{2\pi} - b$

desarrollador predecir si: $V^S \geq 0$

$\hookrightarrow \omega + sH - \frac{\beta s^2}{2\pi} - b \geq 0 \rightarrow \omega \geq b - sH + \frac{\beta s^2}{2\pi}$

$V^S \geq 0 \iff \omega \geq T^S$

Umbral con delegación: $V^D = \omega + (1-\lambda)sH + \lambda a_2(A) - k(a,s) - r_D - b - \frac{\rho}{2} \left[\frac{(1-\lambda)^2 s^2}{\pi} + \sigma_D^2(a,s,A) \right]$

delegación es viable si: $V^D \geq 0$

$\hookrightarrow \omega \geq (1-\lambda)sH + \lambda a_2(A) - k(a,s) - r_D - b - \frac{\rho}{2} \left[\frac{(1-\lambda)^2 s^2}{\pi} + \sigma_D^2(a,s,A) \right]$

$V^D \geq 0 \iff \omega \geq T^D$

Para un lenguaje desconocido: $\gamma^S - r_C \leq 0$

por ende: $T^C \geq T^S$

$T^1 = \min \{T^S, T^C\} = T^S$

Beneficio de la delegación como la reducción del umbral: $\beta = T^S - T^D$

$\hookrightarrow \beta = (b - sH + \frac{\beta s^2}{2\pi}) - [b - (1-\lambda)sH - \lambda a_2(A) + k(a,s) - r_D + \frac{\rho}{2} \left(\frac{(1-\lambda)^2 s^2}{\pi} + \sigma_D^2 \right)]$

$\hookrightarrow \beta = \lambda [a_2(A) - sH] - k(a,s) - r_D + \frac{\rho}{2} \left[\frac{(2\lambda - \lambda^2) s^2}{\pi} - \sigma_D^2(a,s,A) \right]$

si $\beta \geq 0 \rightarrow T^S \geq T^D$

nuevo umbral: $T^2 = \min \{T^1, T^D\}$

$T^2 = T^D < T^S = T^1$ $z^1 = 1 \mid \omega \geq T^S$

$z^2 = 1 \mid \omega \geq T^D$

consideramos 3 regiones:

- $\omega < T^D$, no alcanza ningún umbral $z^2 = 0, z^1 = 0$
 $z^2 - z^1 = 0$
- $T^D \leq \omega < T^S$ $z^2 = 1, z^1 = 0$ $z^2 - z^1 = 1$
- $\omega \geq T^S$ $z^2 = 1, z^1 = 1$ $z^2 - z^1 = 0$

Por ende, se cambia la decisión exactamente cuando:

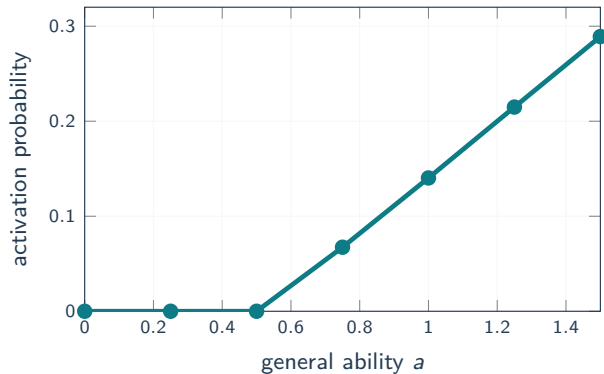
$z^2 - z^1 = 1 \mid T^D \leq \omega < T^S$

Si ω tiene dist. F continua: $E[z^2 - z^1] = Pr(z^2 - z^1 = 1)$

$\rightarrow Pr(T^D \leq \omega < T^S) \rightarrow F(T^S) - F(T^D)$

$\rightarrow Pr(z^2 - z^1 = 1) = F(T^S) - F(T^D)$

8. A numerical stress test exposes the switch



Transparent illustration

- ▶ standard-normal ω
- ▶ parameters fixed except a
- ▶ Assumption 1 verified
- ▶ exact endpoint assertions

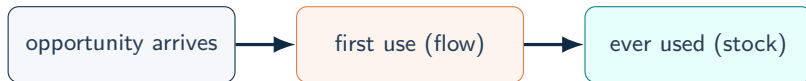
Nothing happens until $B > 0$. This is a mechanism check, not an estimate or calibration.

Source: Repository analysis/activation_band.py and generated CSV.

9. From a first-use flow to a cumulative stock

For an initially unfamiliar language with first-use hazard p_{ik}^g ,

$$\Delta C_i(s) = \sum_{k \in \mathcal{U}_i} \left[(1 - p_{ik}^1)^{s+1} - (1 - p_{ik}^2)^{s+1} \right] \geq 0 \quad \text{if } p_{ik}^2 \geq p_{ik}^1.$$



Strict increase and strict discrete concavity do *not* follow from $p^2 \geq p^1$ alone. A sufficient nondegenerate benchmark is $p^1 = 0 < p^2 < 1$; if $p^2 = 1$, the stock jumps once and then is flat.

Source: Quispe and Xu, Proposition 3 and Appendix proof.

10. Heterogeneity and repositories: extra assumptions

Proposition 4

$$\mathbb{E}[E_i \mid a_i, U_i] = U_i p_i(a_i, A).$$

- ▶ comparable unfamiliar candidates
- ▶ p_i increasing in ability under verification technology
- ▶ strictness needs positive density at threshold

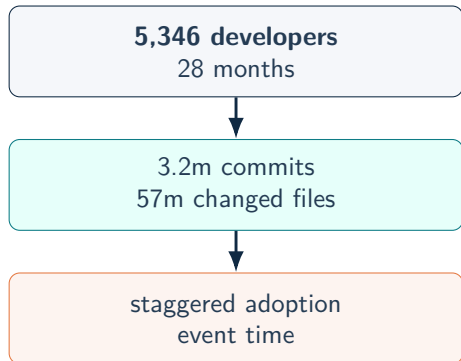
Proposition 5

Repository entry expands if:

- ▶ each repository requires a language
- ▶ its entry cost weakly falls when that language activates
- ▶ strictness needs repositories in the band

Source: Quispe and Xu, Assumptions 2–3 and Propositions 4–5.

11. Evidence: large event study, cautious interpretation



Adoption-month associations

- ▶ active languages: +2.5 vs. 0.9 baseline
- ▶ newly used languages: about +1.2 vs. 0.31 baseline
- ▶ entropy: +0.38
- ▶ cumulative breadth: about +1.6

An unfamiliar-language project may cause both adoption and first use. The authors do not claim definitive causality.

Source: Quispe and Xu, abstract, Sections 2 and 4–5, Tables/Figures reported in text.

12. Same AI, opposite conclusions: one modeling fork

Quispe–Xu

$$\{S, C\} \longrightarrow \{S, C, D\}$$

AI creates an optional delegation mode. General ability supports specification and verification; interaction can later generate information signals. **Prediction:** lower language-entry thresholds, frontier expansion.

Aouad–Lykouris–Zhong

$$x = s + e + a$$

AI assistance, skill, and effort are perfect substitutes. More a crowds out effort; effort governs future skill transitions. **Prediction:** deskilling and possible long-run productivity decline.

The core contrast is menu expansion versus input substitution. Information is secondary in Quispe–Xu.

Source: Quispe and Xu, Section 3 and learning appendix; Aouad, Lykouris, and Zhong, Sections 2–4.

13. Lean: paper equation $\rightarrow$ proof $\rightarrow$ meaning $\rightarrow$ check

Paper equation

$$Z^2 - Z^1 = \mathbf{1}\{T^D \leq \omega < T^S\} \quad (T^D < T^S).$$

Lean statement and proof

```
theorem activationBand : activationBandSpec := by
  ...
  have hcopilot :
    Tsolo <= Tsolo - (gamma * skill - interactionCost) := by linarith
  have hdelegate : Tdelegate < Tsolo := by linarith
  rw [min_eq_left hcopilot, min_eq_right hdelegate.le]
  by_cases hD : Tdelegate <= omega
  by_cases hS : Tsolo <= omega
  simp [hD, hS, not_lt.mpr hS]
```

Explanation

The draft represents activity with integer-valued indicators and splits on both cutoffs. The paper-scoped checker did not verify the draft: the agent exhausted its usage limit before EconCSLib generated the paper folder.

Build/check result **FAIL: no paper folder; missing status.json.**

Source: Own GPT-5.6 Sol xhigh attempt; exact commands and outputs in LEAN_STATUS.md. No lean/ folder was generated.

14. Verdict

What the model earns

- ▶ weak expansion from menu inclusion
- ▶ strict activation in a precisely defined band
- ▶ testable extensive-margin heterogeneity

What the evidence does not earn

- ▶ causal identification of the agent mechanism
- ▶ proof that activated languages become human skill

The conclusion flips when AI changes from a **substitutable input** into an **optional delegated mode**.

Source: Synthesis of [arXiv:2605.25438v2](#) and [arXiv:2605.11350](#).